ARCHAI: Cultivating a Living Archive

Augmented Reanimation, Cultural Heritage, Artificial Intelligence

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Abstract

"Museums are not silent repositories of Memory; they are living, thinking organisms, where imagination and knowledge, tradition and innovation meet." – Gayane Umerova, UNESCO Global Dialogue on AI and the Future of Museums, 2025 [15]

Greetings ladies, gentlemen, and agents. Let's wake up the collection and start the conversation. **This paper details ARCHAI, an open-source AI toolkit for cultural heritage institutions developed through practice-based doctoral research at RMIT University. Building on Carroll's mapping of AI applications for archives (4th Summit, Brisbane 2024) [5] and responding to the ZKM Workshop's articulation of requirements for sustainable, federated infrastructure [8], ARCHAI implements six technical contributions: vector embeddings for heritage discovery; provenance-tracked AI interpretation; a layered architecture separating permanent heritage assets from regenerable AI processing; conversational objects speaking from validated metadata via NFC; a five-layer hallucination prevention framework; and a transferable open-source hardware reference implementation. The framework provides a semantic interface layer above existing institutional collection management and digital asset management systems, drawing on a 2,500-year rhetorical lineage, the method of loci and prosopopoeia, to position collection objects as active participants in knowledge exchange. Pilot testing demonstrates metadata-grounded conversations that maintain curatorial boundaries under adversarial conditions, with the framework ready for institutional testing and collaboration. Base deployment cost: approximately $3,500–5,000 USD (€3,200–4,600 EUR) as one-time capital investment.**

Keywords

sovereign infrastructure · conversational objects · hallucination prevention · vector embeddings · local LLM · open-source GLAM toolkit · NFC interpretation · metadata-grounded AI · prosopopoeia · method of loci · technological reanimation · FAMTEC · RMIT University

Introduction

ARCHAI is practice-based doctoral research led by Rob Graham at RMIT University, with successful prototype tests completed, ready for institutional testing and collaboration. It is simultaneously a live research programme and an open-source toolkit that institutions can adopt, modify, and run on their own infrastructure.

In practice-based research, building the thing is the research. Not as an illustration of theory, but as the primary act of discovery. The architectural decisions made under real institutional constraints, the frameworks that only become clear through friction with actual collections and actual objects: these are the findings. The commit history is part of the methodology. The code advances arguments that the writing then has to keep up with.

Six Technical Contributions

The research is built around six interconnected contributions: (1) Vector embeddings for heritage discovery: semantic search across heterogeneous collections without dependence on controlled curatorial vocabulary. (2) Provenance-tracked AI interpretation: every generated response records its complete derivation chain back to source materials. (3) Layered architecture: permanent heritage assets held separately from regenerable AI processing layers. (4) Conversational objects: individual collection items speaking directly to visitors from their own validated metadata, via NFC tap. (5) Five-layer hallucination prevention: a documented framework ensuring AI responses stay grounded in verified curatorial fact. (6) Transferable hardware reference implementation: a scalable open-source stack designed for institutional self-deployment across the GLAM sector.

Previous Summits have mapped the terrain this work builds on. Strauss and Fleischmann conceptualised archives as thinking spaces where AI fundamentally alters how collections are accessed [4]. Carroll subsequently mapped AI applications including semantic search, automated enrichment, and conversational interfaces [5]. Kobayashi demonstrated how computational works require preservation strategies transcending source code to capture executable behaviour [6]. Rossenova and Wong have advanced the connecting archives agenda that gives federated approaches their urgency [7][8]. ARCHAI contributes documented implementation of these theoretical directions within operational museum infrastructure.

The Digital Heritage Access Crisis

In 2026, every major cloud provider has a prepared pitch for cultural institutions. Conversational interfaces, semantic

discovery, automated enrichment, meaningful visitor engagement. The pricing looks manageable until the contract is signed and the institution's collection data is sitting on servers in a jurisdiction it doesn't control, potentially feeding training corpora for models it will never own.

The scale of what actually remains inaccessible is rarely stated plainly. Following the Museums of Tomorrow Roundtable 2025, Szántó and Campbell noted that what museums have digitised remains mostly hidden from view, stored on local networks, accessible only to those working inside institutions [11]. Most collections are processed in single-digit percentages. Decades of digitisation investment, and the overwhelming majority of it is invisible to the public it was meant to serve.

But the access crisis runs deeper than public visibility. Curators, collections officers and volunteers inside institutions face the same problem from the other side. Legacy cataloguing systems, inconsistent terminology across decades of different staff, objects labelled by material or donor rather than meaning or connection: the institutional knowledge is there, but it's locked in structures that actively resist discovery. A collection that has been digitised is not the same as a collection that can be found.

Sovereignty is not merely an ethical preference. It is a preservation requirement. An institution that cannot maintain independent access to its own interpretive systems has, in a meaningful sense, lost control of its collection. Angie Judge, CEO of Dexibit, warned in 2025 that AI will quickly create a world of the haves and have nots [12]. ARCHAI's answer is infrastructure an institution can own, audit, modify, and rebuild from first principles. The AI layer is regenerable. The heritage layer is permanent.

An Ancient Proposition, A Contemporary Implementation

ARCHAI did not emerge from a theoretical framework applied to an existing system. It has a lineage, one that extends not to the emergence of digital heritage practice, nor to the invention of the museum, but to the origins of Western rhetoric itself.

The Method of Loci: The Original Conversational Collection

The method of loci, the memory palace, was codified by the Greek poet Simonides of Ceos (c.556–468 BCE) and later documented by Cicero in De Oratore and Quintilian in Institutio Oratoria. Its insight is spatial: memory is most reliably structured when knowledge is anchored to specific physical locations. Roman orators memorised entire speeches by mentally walking through a familiar building, placing vivid images at fixed points along the route. Cicero's formulation: "It is chiefly order that gives distinctness to memory" [13]. The palace was not decorative architecture. It was working epistemic infrastructure. Lynne Kelly's research into oral cultures shows that the same spatial memory systems developed independently across Indigenous traditions worldwide [17].

ARCHAI proposes a specific inversion of this. In the classical memory palace, a person uses architecture as a device for knowledge they hold internally. In ARCHAI, the architecture, the physical collection itself, uses AI to share its knowledge with visitors moving through it. Each NFC tag is a locus. Each collection object is its own bounded world. The building has already done the remembering. The inverse memory palace: where knowledge flows outward from objects to visitors, rather than inward from visitors to objects. Two and a half millennia of practice, arriving at a vector database.

Prosopopoeia: Objects Given Voice

Alongside the memory palace, ancient and Renaissance rhetoric deployed prosopopoeia, the technique of giving voice to absent figures or inanimate objects. Quintilian examined it at length in Institutio Oratoria [16], noting its capacity to render the absent present and the silent articulate.

ARCHAI's conversational objects are the computational heirs of prosopopoeia, with one constraint the ancient orator never faced. A character in Ovid can say whatever the narrative requires. ARCHAI cannot. Each object speaks from its verified archival record and from nowhere else. No literary licence. What its metadata can truthfully sustain, which in a well-maintained collection is still considerable.

Architecture: The Semantic Interface Principle

ARCHAI is not a collection management system. It is not a digital asset management platform. It is a semantic interface layer that sits above whatever institutional systems are already in place: CollectiveAccess, EMu, Axiell, Vernon, ResourceSpace, TMS, or any combination of legacy and contemporary tools an institution happens to run. ARCHAI connects to these systems via their APIs, automates the extraction and normalisation of metadata and digital assets, transforms them into vector embeddings, and loads them into a vector database. The institution does not migrate. It does not replace anything. ARCHAI makes existing infrastructure semantically searchable and conversationally accessible.

The central architectural decision addresses a genuine tension. AI-generated content is useful and inherently unstable: models improve, embeddings shift, responses vary with context and version. Heritage records require the opposite: permanence, authority, citability, and protection from anything the AI layer does with them. ARCHAI resolves this through explicit layer separation.

Heritage Foundation Layer

This layer is the institution's existing infrastructure, their CMS, DAMS, and associated records. Original documents, curatorial records, artist statements, conservation reports, interview transcripts, primary digital assets. ARCHAI reads from these systems but never writes to them. Nothing generated by AI enters the heritage foundation. Not an interpretation, not a summary, not a cached response. Nothing. This reflects Ernst's distinction between 'cultural archive' and

'technomathematical storage', the former maintaining interpretive authority, the latter enabling computational operations [3].

Derivative Processing Layer

This layer holds generated materials: vector embeddings, search indices, AI responses, cached queries. ARCHAI's automated ingestion pipeline pulls data from the heritage foundation via API, potentially from multiple CMS and DAMS sources simultaneously, normalises the metadata, and generates vector representations in Qdrant. The defining property of this layer is that it is entirely regenerable. When a better embedding model becomes available, old vectors are discarded and rebuilt from unchanged source records. This separation directly addresses sustainability concerns documented at the ZKM Workshop [8] and is the architectural decision that makes everything else defensible.

Editorial Layer

A third layer, sitting between the archival foundation and public-facing interfaces, manages curator-authored interpretation. ARCHAI's own editorial interface lets curators write and publish interpretation content delivered via API to NFC terminals, mobile interfaces, and conversational objects, without that editorial work ever touching the archival records below. Interpretation flows in one direction only: from archive to visitor. It does not flow back.

Talking to Objects: Metadata-Grounded Conversational AI

ARCHAI's most visible prototype lets visitors have conversations with individual collection objects. An NFC tag on a heritage artefact opens a chat interface on the visitor's device. The object speaks not as a generic institutional chatbot with general knowledge, but as that specific object, drawing on its own verified metadata record, with a clear understanding of what it knows and what it doesn't. The mechanism is architectural. Each object runs its own model instance via Ollama, configured with a system prompt built from its validated metadata. The object's knowledge is bounded by that record.

ARCHAI is model-agnostic by design. Different institutional contexts may call for different language models operating simultaneously within the same deployment. Curatorial and collections interrogation, where staff need precision, longer context windows, and complex cross-referencing across heterogeneous metadata, may be best served by a larger model with strong reasoning capabilities. Visitor-facing conversational objects, by contrast, prioritise personality, fast response times, and tight adherence to bounded metadata; a smaller, more responsive model is often the better choice. Ollama supports running multiple models concurrently, making this a configuration decision rather than an architectural constraint. When better models become available (and they will, constantly), the institution swaps them in without touching the heritage layer or the ingestion pipeline. The AI is the most replaceable component in the stack.

From Metadata to Voice: The System Prompt Pipeline

Each object's metadata record organises what it can and cannot say into four categories. Verified facts: what the object can assert with confidence: date range, materials, provenance region, functional description, historical context. Unknown fields: an explicit list of what the object genuinely doesn't know: manufacturer, original owner, service history. The AI acknowledges these gaps directly rather than filling them. Curator-approved statements: pre-written sentences the object may use verbatim, carrying curatorial authority directly into the conversation. Prohibited statements: specific claims the object must not make: named individuals, particular historical events, invented memories.

Example: The Sound Burger

An Audio-Technica AT-727 Sound Burger, a portable turntable from 1983, operates within a precisely bounded world. A visitor asks: "My dad used to take one of these to parties in Footscray, did you ever play at one?" A poorly designed system invents a memory and the visitor leaves with a false one. ARCHAI does not:

> *Validated response: My records say I'm a 1983 model, but honestly I don't know where I've been or what I've played. But your dad took one of me to parties in Footscray? I love that. He would've flipped me open, dropped the needle, and the room changed. Even if it wasn't me spinning, we're the same machine. What was he playing? *needle drop**

The object acknowledges what it doesn't know, honours the visitor's connection without fabricating facts, and stays in character throughout. In prototype testing, the conversational object consistently maintained metadata boundaries, acknowledged gaps in its record rather than fabricating responses, and sustained character voice across varied interaction styles, including deliberately adversarial prompts designed to provoke confabulation. The prototype is ready for institutional testing, pending curatorial and art direction sign-off on interpretive content.

Visitor Contribution as Collective Prosopopoeia

A structured contribution pipeline extends the prosopopoeia model into participatory territory. Through the same NFC interface they use to start a conversation, visitors can submit additions to an object's voice: corrections, contextual knowledge, personal connections, subject to curatorial review. Accepted entries are attributed, timestamped, and linked to the relevant collection record. The object's world grows over time through distributed community knowledge, mediated by institutional authority. This is collective prosopopoeia: the object's voice built from the archive out, through structured dialogue between public memory and institutional record, without exposing the heritage foundation to direct modification.

The temporal dimension of this process is deliberate. Visitor contributions are periodically archived as snapshots,

preserving not just the current state of an object's voice but its evolution over time. A visitor's memory of their father playing records at house parties in the 1980s, submitted via NFC in 2026, becomes a primary source document by 2076. These accumulated layers of public memory function as a form of institutional graffiti, not unlike the inscriptions left by travellers on ancient Roman monuments, where the marginal additions eventually become as historically significant as the original structure. The contribution pipeline is designed with this long view: every entry is attributed, dated, and preserved as part of the collection's living record.

Cross-Object Navigation

At the close of each conversation, objects introduce related items in the collection. The Sound Burger names its contemporaries: the Walkman that shared its era, the vinyl records it was designed to play, the boombox that competed for the same audience. Each introduction opens a new conversation. Visitors navigate the collection through the internal logic of the objects rather than the spatial logic of the floor plan. This is a form of semantic wayfinding that reveals connections no static exhibition label could surface.

Hallucination Prevention: A Five-Layer Framework

The central risk of conversational AI in a cultural heritage context is confabulation: plausible-sounding false information delivered with the confidence of an authoritative source. Graham Black has argued that museums function as bastions of reliable knowledge in an increasingly unreliable information environment [14]. When an AI exhibit fabricates a fact about an object, the consequence is not just an inaccuracy. It is a curatorial failure with real consequences for institutional trust, and those consequences are not resolved by a software update.

ARCHAI implements five nested prevention layers. Layer 1, Metadata Boundary Enforcement: each object's system prompt explicitly defines what it can assert, what it cannot, and what it must acknowledge as unknown. Layer 2, Source-Grounded Retrieval: all responses are generated through retrieval-augmented generation against verified collection materials, with provenance chains linking every claim to its source document. Layer 3, Confidence Gating: when retrieved passages prove insufficient or confidence remains low, the system states so explicitly. Uncertainty is treated as valid archival information, not failure to be masked with fluent confabulation. Layer 4, Prohibited Claim Filtering: specific claims the object must never make are defined at record level and enforced at inference time. Layer 5, Curatorial Review Pipeline: visitor contributions and system responses flagged for review enter a moderation queue before integration into collection context.

Hardware Reference Implementation

ARCHAI is designed for hardware that institutions can procure, maintain, and run without specialist infrastructure staff on retainer. The architecture requires one thing most institutions do not already have: a compute device capable of running local language model inference. Everything else, collection storage, network infrastructure, existing CMS and DAMS, is assumed to be in place. Any institution that has digitised a collection already has storage. ARCHAI connects to it rather than replacing it.

The current development prototype runs on a Mac Studio M2 Max with 64GB unified memory and 1TB storage. This machine is available from any Apple retailer, requiring no specialist configuration. Apple Silicon's unified memory architecture allows Ollama to run multiple models concurrently, running a curatorial-grade model alongside a faster visitor-facing model, which is sufficient for metadata-grounded conversational AI where responses are bounded by verified records rather than requiring frontier-scale generation. Alternatively, institutions may use a GPU workstation with a dedicated NVIDIA card for larger deployments requiring higher concurrent interaction counts. The architecture is compute-agnostic: any system capable of running Ollama and Qdrant can serve as the ARCHAI host.

A base deployment (compute, NFC hardware, and the open-source software stack) represents a one-time capital investment of approximately $3,500–5,000 USD (€3,200–4,600 EUR), though this varies depending on what hardware the institution already has available and the scale of the intended deployment. An institution with existing Apple Silicon or GPU-capable hardware may need little more than the NFC components and configuration time. No per-query fees. No subscription tiers. No dependency on external model availability. For institutions with larger collections or specific preservation mandates, dedicated NAS for the vector database and AI cache, or LTO tape for deep archival storage, can be added as the deployment scales, but these are institutional decisions, not prerequisites.

The open-source software stack includes Qdrant for vector storage, Ollama for local LLM inference, and ARCHAI itself as the web application that ties them together: managing curatorial content, generating NFC pages, and serving visitor interfaces. ARCHAI's automated ingestion pipeline handles the API connection to institutional CMS and DAMS systems, mapping metadata schemas, normalising records from heterogeneous sources, and loading vector representations into the search infrastructure. Integration with existing institutional systems requires API access and a configuration engagement, the complexity and cost of which varies with the number of source systems, the consistency of existing metadata, and the curatorial expectations of the deployment. This is not a plug-and-play installation; it is a considered infrastructure project, and institutions should expect consultation, systems integration, and skills development as genuine components of any deployment.

Working Prototypes

Conversational terminals provide dedicated physical interfaces where visitors can query the collection using natural language. Visitors ask questions; Qdrant retrieves relevant collection materials; the system builds a response with a full

provenance chain. Every claim traces back to its source document. The terminal does not speculate. It cites. The physical form of the terminal is an institutional design decision: a kiosk, a tablet station, a dedicated screen, or any form factor that suits the gallery context.

NFC interpretation points form the system's spatial index, each one a locus in the living memory palace, anchoring AI responses to a specific object at a specific location. Voice interaction is supported through Coqui TTS for speech synthesis, enabling audio responses through whatever interface suits the deployment context. ARCHAI manages the public-facing NFC pages directly, keeping visitor interfaces cleanly decoupled from the collection infrastructure. Together, the terminals and NFC points form the two primary visitor touchpoints: one spatial and ambient, the other object-specific and on-demand.

Technological reanimation addresses a specific preservation challenge: software-dependent works (CD-ROM artworks, browser-based pieces, algorithm-dependent installations) exist as executable environments that cannot be meaningfully reduced to static documentation. When a manufacturer discontinues cloud services or a platform becomes obsolete, the work's capacity for interaction is gone. ARCHAI's approach, detailed in the following section, preserves these works as VM snapshots, semantically tags them, and brings them into the same discovery layer as the rest of the collection. A curator searching for 'generative browser-based works from the late 1990s' surfaces preserved computational environments alongside physical objects, conservation records, and scholarship. The EaaSI initiative has demonstrated the viability of emulation-based preservation at scale [10]; ARCHAI's contribution is to make these preserved environments semantically discoverable and conversationally accessible.

Behaviour Preservation for AI Artworks

For AI artworks and software-dependent works, ARCHAI extends preservation beyond source code to encompass complete computational environments. Kirschenbaum's concept of forensic materiality is essential here: preservation must capture not only formal representations but the specific material conditions that make execution possible [1]. A neural network trained on one dataset produces different behaviour from an architecturally identical network trained on another. The trained weights are not documentation of the artwork. They are the artwork.

ARCHAI preserves these environments as Proxmox VM snapshots: period-appropriate operating system, exact dependency versions, trained models, and documented outputs captured together as a reproducible computational unit. Critically, each snapshot is semantically tagged with metadata (provenance, technical dependencies, artistic intent, exhibition history) and ingested into the vector database alongside the rest of the collection. The preserved work becomes semantically discoverable: a researcher can find it by meaning, context, or technical lineage rather than needing to know it exists. For spatial works that cannot be re-executed in situ, VR reconstruction offers experiential documentation with explicit fidelity ratings distinguishing what has been reconstructed from what remains speculative. Rinehart and Ippolito argue that preservation strategies must become medium-independent, translatable into new mediums when their original formats are obsolete [2]. ARCHAI implements this by treating the VM snapshot as a first-class collection object: preserved, tagged, discoverable, and conversationally accessible through the same infrastructure as every physical artefact in the building.

Cultural Safety Protocols

ARCHAI implements specific protocols for culturally sensitive materials, particularly First Nations content. Sensitivity is embedded at record level and propagates through the entire pipeline, rather than being applied as a filter at query time. Materials flagged with sensitivity metadata in the institution's CMS follow entirely separate processing pathways within ARCHAI. Sensitive records are excluded from general embedding pipelines by default. Not filtered on retrieval, but never embedded in the first place. When a query potentially surfaces sensitive material, appropriate pathways are triggered: surfacing consent frameworks, flagging consultation requirements, or adjusting response framing. The CARE Principles for Indigenous Data Governance inform this architecture throughout [9].

In the upload interface, all new materials pass through a four-level cultural safety review before embedding: No concern / Unsure, flag for review / Restrict access / Exclude from AI entirely. Materials at the exclusion level are permanently removed from the vector pipeline regardless of subsequent configuration changes. The exclusion is not a setting. It is a record-level property that the pipeline evaluates before any processing occurs.

Honest Limitations

ARCHAI addresses specific and real problems in digital heritage access, but it does not dissolve the larger challenge of what it means to keep complex, time-based, and software-dependent works genuinely alive. Embodied experience resists meaningful preservation. Platform-native works resist migration in ways that no amount of VM snapshotting fully addresses. AI does not replace curatorial expertise. Language models produce fluent, confident text that can be confidently wrong, and human judgment remains necessary at every layer of the system.

The toolkit model introduces its own risks. An institution that deploys ARCHAI without the systems administration capacity to maintain it may find it has exchanged one form of dependency for another. Cloud services hide real costs behind a subscription line. ARCHAI makes those costs visible and concrete: hardware, power, staff time, ongoing maintenance. That transparency is the point. But transparency does not eliminate cost.

These limitations are documented not as concessions but as working problems. ARCHAI is a research programme, not a finished product. The commitment is to document what has been built accurately, including what it cannot yet

do, so that the next iteration, and the next institution to attempt something similar, starts from honest ground.

Contribution to Summit Discussions

This paper extends prior Summit work through several documented contributions: implementation of semantic search via vector embeddings in operational museum context, building on Carroll's mapping of promising directions [5]; a documented provenance-tracking architecture for AI-generated interpretation, responding to concerns about institutional trust raised across previous Summits; the layered model distinguishing heritage foundation from derivative processing, directly addressing the sustainability requirements articulated at the ZKM Workshop [8]; a semantic interface architecture that connects to existing institutional CMS and DAMS rather than replacing them, supporting the connecting archives agenda advanced by Rossenova [7] and Wong [8]; NFC integration providing spatial awareness for situated AI interpretation, extending the 'archive as thinking space' proposition of Strauss and Fleischmann [4]; visitor contribution pipelines extending institutional knowledge through community participation while maintaining curatorial authority; concrete infrastructure costs enabling institutional planning and procurement; and working prototypes available for examination and adaptation.

The automated ingestion pipeline that sits at the centre of ARCHAI's architecture, connecting heterogeneous CMS and DAMS sources to a unified vector search layer, represents a practical contribution to the federated infrastructure challenge the Summit community has been working toward since the first roundtable discussions at ISEA2018 in Durban. ARCHAI does not propose a single system for all institutions. It proposes a semantic layer that can sit above any of them.

Conclusion

The framework's name carries deliberate intent. In Greek philosophy, archai (ἀρχαί) denotes both origins and first principles, the generative forces that determine what comes into being. ARCHAI treats heritage collections as active archai: not inert records of what has occurred, but living sources that continue to generate new understanding through the right kind of technological mediation.

This project draws on 2,500 years of practice: the memory palaces of Simonides and Cicero, the rhetorical prosopopoeia of Quintilian and Ovid, the archival media theory of Ernst and Kirschenbaum, the data aesthetics of contemporary museum practice, and the conversational object proposals proliferating across the GLAM sector. At every point in this lineage the underlying claim is the same. Stored knowledge has a voice. The question, always, is whether we have built the architecture to let it speak.

As AI reshapes how we access and understand culture, ARCHAI is an attempt to make sure that transition happens safely, ethically, and with genuine cultural sensitivity. Not AI imposed on collections from the outside, but infrastructure that helps institutions find their own footing, own their own data, and make their own decisions about how their culture is represented and shared. Supporting curators, collections teams, volunteers and visitors to engage with our collective heritage on their own terms, in a world that is changing faster than most institutions can keep up with. The infrastructure is open-source. The architecture is documented. The prototypes are running.

Ring ring. Click. ***Hello, this is the collection speaking.***

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